A Closed-Loop Subspace–RFS Framework for DOA Estimation and Multi-Target Tracking

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***Abstract*—We present a closed-loop subspace–RFS framework that couples direction-of-arrival (DOA) estimation with random-finite-set (RFS) multi-target tracking and, unlike prior open-loop cascades, analyzes the coupling itself. The closed loop is *front-end-agnostic*: the tracker's temporal prediction is fed back to refine the signal subspace of any subspace DOA estimator. We instantiate it with a higher-order-cumulant front-end (COP, which extends operation to the underdetermined regime $K > M-1$; the resulting tracker is COP-RFS) and a covariance front-end (MUSIC). Our central contributions are theoretical: (i) a posterior Cramér–Rao bound for the *complete closed-loop* estimator–tracker, proving the temporal prior restores identifiability exactly where the data-only bound diverges; and (ii) a recasting of the subspace refinement as minimum-variance inverse-variance fusion on the Grassmann manifold, with a validation gate guaranteeing the loop never degrades performance. The framework is also *back-end-agnostic*, integrating track-oriented PHD, labeled multi-Bernoulli (LMB), and δ-GLMB filters; a physics-based constant-acceleration model makes the δ-GLMB back-end best for maneuvering targets, yielding our recommended configuration—a gated closed-loop front-end (COP, or the lower-variance MUSIC when determined) driving a physics-based, interchangeable SOTA back-end (here, δ-GLMB). Monte-Carlo experiments (up to 500 trials, 95% confidence intervals) show that COP-RFS detects significantly more targets than a MUSIC–PHD baseline when $K > M-1$, and that the gated loop lowers low-SNR localization error by up to 56% (COP) and 44% (MUSIC) while cutting moving-target identity switches by 15%, matching the open-loop baseline otherwise. We delineate the operating regime in which the higher-order front-end is advantageous rather than claiming uniform superiority.**

***Index Terms*—Direction-of-arrival estimation, high-resolution, higher-order cumulants, underdetermined systems, multi-target tracking, probability hypothesis density filter, random finite sets, constrained optimization.**

I. Introduction

High-resolution direction-of-arrival (DOA) estimation is a fundamental problem in array signal processing with critical applications in radar, sonar, communications, and defense systems [2], [3]. Classical high-resolution subspace methods such as MUSIC [4], ESPRIT [5], and Capon's MVDR beamformer [6] achieve super-resolution beyond the Rayleigh limit but are restricted to the *overdetermined* case where the number of sources K does not exceed $M-1$, the number of sensors minus one. This constraint poses a significant challenge in modern defense scenarios where a sensor array may face more incoming threats than its aperture can nominally resolve.

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Higher-order statistics (HOS) offer a principled approach to overcome this limitation [7], [8]. By exploiting the 2ρth-order cumulant structure of received signals, Choi and Yoo [1] proposed a high-resolution Subspace Constrained Optimization of the Pseudo-spectrum (COP) algorithm, which extends the effective array aperture through a *virtual array* of size $M_v = \rho(M-1)+1$, thereby resolving up to $\rho(M-1)$ sources—far exceeding the classical $M-1$ limit. Alternative approaches to the underdetermined DOA problem include sparse array designs such as nested arrays [9] and coprime arrays [10]–[12], as well as sparsity-based and sparse-Bayesian-learning methods [13]–[15]. However, these approaches either require specialized array geometries or assume signal sparsity, whereas COP operates on standard ULAs and exploits higher-order cumulant structure. A key advantage of higher-order cumulants is the automatic suppression of Gaussian noise, since all cumulants of order greater than two vanish for Gaussian processes [16]—a robustness that holds even for spatially-colored, unknown noise. These strengths come at the cost of higher estimation variance; a central premise of this work is that embedding HOS *within a closed loop* turns that liability into a net gain, as the tracker's temporal prior supplies exactly the variance reduction that HOS lacks.

While the COP algorithm addresses the static underdetermined DOA problem, practical defense applications require *multi-target tracking*: maintaining continuous trajectories across successive radar scans even as targets maneuver, appear, or cross. Existing DOA-tracking pipelines typically chain a covariance-based estimator (MUSIC, ESPRIT) with a Bayesian tracker (Kalman, PHD) in an *open-loop* fashion, where the tracker consumes DOA estimates but provides no feedback to the estimator. This design has two weaknesses: (i) the estimator cannot resolve $K > M-1$ sources, and (ii) the tracker receives no help from higher-order signal structure.

The GM-PHD filter [17] provides an efficient approximation for RFS-based multi-target tracking, but the standard predict–update formulation does not maintain explicit track identity. Several extensions have addressed this limitation: Panta et al. [18] introduced data association schemes for the GM-PHD filter, Clark and Vo [19] analyzed convergence properties, and Georgescu and Willett [20] applied the GM-PHD to phased array DOA tracking. More fundamentally, the labeled multi-Bernoulli (LMB) and δ-GLMB filters [21], [22] solve the track identity problem within the RFS-theoretic framework itself. However, none of these works exploit higher-order cumulant structure for the DOA estimation front-end, nor do they establish a closed-loop feedback between tracker predictions and the estimation subspace.

This paper proposes a *gated closed-loop subspace–RFS* *framework* (Fig. 1) that, unlike prior open-loop cascades, makes the estimator–tracker *coupling* the object of design and analysis: a subspace DOA front-end feeds a random-finite-set (RFS) tracker whose temporal prediction is fed back—through a two-stage validation gate and a Temporal-COP (T-COP) refinement—to sharpen the estimator. The framework is deliberately modular: *front-end-agnostic* (higher-order-cumulant COP, yielding COP-RFS for the underdetermined regime $K > M-1$, or covariance MUSIC) and *back-end-agnostic* (PHD, LMB, or δ-GLMB). Its *recommended configuration*—the destination of this paper—is a gated closed-loop front-end (COP, or MUSIC when determined) driving a *physics-based*, interchangeable SOTA back-end (δ-GLMB), robust across source count, SNR, motion, and signal type. The principal contributions are:

Contribution 1 (Theory): A posterior CRB for the complete closed loop. We derive the first Cramér–Rao bound for the coupled higher-order-cumulant/RFS tracker (Theorem 1), via the posterior-CRB recursion. It proves that the temporal prior strictly tightens the bound in the Löwner order and, crucially, that it *restores identifiability*—keeping the bound finite—precisely in the underdetermined regime where the data-only bound diverges. It is an honest, closed-loop analysis rather than an unsupported "exact" static-cumulant CRLB.

Contribution 2 (Theory): Provably minimum-variance T-COP refinement. We recast tracker-aided T-COP subspace refinement as a basis-invariant inverse-variance fusion on the Grassmann manifold and prove it is minimum-variance (Theorem 2), with an explicit prior-accuracy condition $\|\mathbf{b}\|^2 < D(\sigma_e^2 + \sigma_p^2)$ under which it strictly reduces estimation error. A validation gate enforces this condition, so the closed loop *never degrades* performance—improving when the prior is accurate (low SNR, slow targets) and falling back to open-loop COP otherwise. It thereby avoids a gauge-dependent linear interpolation, which is not a well-defined subspace operation.

Contribution 3 (Practice): Underdetermined capability and honest evaluation. A physics-based track-oriented PHD (TO-PHD) back-end preserves track identity through crossings via predict–identify–update association. Through Monte-Carlo experiments (up to 500 trials) with 95% confidence intervals, and against MUSIC and closed-loop-MUSIC baselines, we show that COP-RFS detects significantly more targets than the cascade baseline when $K > M-1$, and we *delineate the operating regime* in which the higher-order front-end is advantageous rather than claiming uniform superiority. A 2×2 (front-end $\times$ loop) study confirms that the closed-loop refinement improves *both* the COP and MUSIC front-ends, establishing it as a front-end-agnostic principle rather than a COP-specific trick. A tracking back-end ablation (standard GM-PHD, our TO-PHD, the labeled multi-Bernoulli (LMB) filter, and the full δ-GLMB filter) further shows the framework is *back-end-agnostic*: all three labeled trackers cut identity switches by an order of magnitude over GM-PHD, with the full δ-GLMB giving the cleanest identity, LMB the best aggregate accuracy, and TO-PHD a lightweight alternative. Endowing the δ-GLMB with a physics-based constant-acceleration model makes it best-in-class for *maneuvering* targets, yielding our recommended configuration: a gated closed-loop front-end driving a physics-based, interchangeable SOTA back-end (δ-GLMB). Sequential-deflation COP (SD-COP) is included as a supporting mechanism for the super-underdetermined regime.

The remainder develops each component of this system: the signal/cumulant model (§II) and closed-loop T-COP front-end (§III); the track-oriented PHD back-end (§IV); the CRLB (§V) and the two central theorems (§VI); and the Monte-Carlo evaluation (§VII).

II. Signal Model and Higher-Order Cumulants

A. Array Signal Model

Consider a ULA with M sensors and inter-element spacing $d = \lambda/2$, where λ is the wavelength. In the presence of K narrowband far-field sources impinging from directions $\{\theta_k\}_{k=1}^K$, the received signal vector at snapshot t is

$$\mathbf{x}(t) = \mathbf{A}(\boldsymbol{\theta})\,\mathbf{s}(t) + \mathbf{n}(t), \quad t = 1, \ldots, T \tag{1}$$

where $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \ldots, \mathbf{a}(\theta_K)] \in \mathbb{C}^{M\times K}$ is the steering matrix with

$$\mathbf{a}(\theta) = \begin{bmatrix} 1 & e^{j2\pi d \sin\theta} & \cdots & e^{j2\pi(M-1)d\sin\theta} \end{bmatrix}^T, \tag{2}$$

$\mathbf{s}(t) \in \mathbb{C}^K$ contains the source signals, and $\mathbf{n}(t) \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_M)$ is additive white Gaussian noise. When $K > M-1$, the system (1) is *underdetermined*, and classical subspace methods based on the second-order covariance matrix $\mathbf{R} = \mathbb{E}[\mathbf{x}\mathbf{x}^H]$ cannot resolve all sources.

B. Higher-Order Cumulant Matrix

The 2ρth-order cumulant of the received signal exploits the non-Gaussian structure of source signals to extend the effective aperture [1], [23]. For $\rho = 2$ (fourth-order), the cumulant element is defined as

$$c_4(i_1, i_2, i_3, i_4) = \mathrm{cum}(x_{i_1}, x_{i_2}, x_{i_3}^*, x_{i_4}^*) \tag{3}$$

where $\mathrm{cum}(\cdot)$ denotes the joint cumulant. Using the sum co-array structure, the lag index $\tau = (i_1 + i_2) - (i_3 + i_4)$ ranges over $[0, 2(M-1)]$, yielding a virtual array of size

$$M_v = \rho(M-1) + 1. \tag{4}$$

The cumulant matrix $\mathbf{C}_{2\rho} \in \mathbb{C}^{M_v \times M_v}$ has the Hermitian Toeplitz structure

$$\mathbf{C}_{2\rho} = \mathbf{A}_v(\boldsymbol{\theta})\,\boldsymbol{\Gamma}_s\,\mathbf{A}_v^H(\boldsymbol{\theta}) \tag{5}$$

where $\mathbf{A}_v(\boldsymbol{\theta}) \in \mathbb{C}^{M_v\times K}$ is the *virtual* steering matrix with extended aperture, and $\boldsymbol{\Gamma}_s = \mathrm{diag}(\gamma_1, \ldots, \gamma_K)$ contains the source cumulant powers. The diagonal structure of $\boldsymbol{\Gamma}_s$ in (5)—and hence the rank-K virtual-array model—holds under the standard higher-order-statistics assumptions [7], [8], [23]: the K source signals are (A1) mutually statistically independent, (A2) non-Gaussian, and (A3) possess nonzero 2ρth-order autocumulants $\gamma_k \neq 0$. Under (A1)–(A3) all cross-cumulants between distinct sources vanish, so that the 2ρth-order cumulant tensor, reindexed by the sum-co-array lag $\tau = (i_1{+}i_2) - (i_3{+}i_4)$, collapses to the Hermitian Toeplitz matrix (5) with $\mathbf{A}_v$ of full column rank $K \le M_v - 1$; the detailed tensor-to-matrix reduction is given in [1], [23].

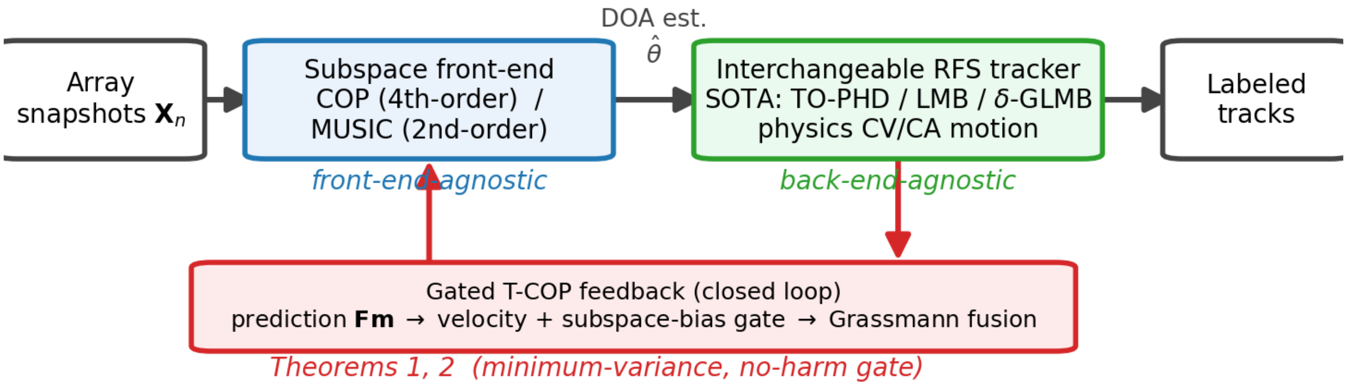

Fig. 1: The proposed gated closed-loop subspace–RFS framework. A front-end-agnostic subspace estimator (COP or MUSIC) feeds a back-end-agnostic RFS tracker (TO-PHD, LMB, or δ-GLMB with physics-based CV/CA motion); the tracker's prediction is fed back through a validation gate and a minimum-variance T-COP refinement (Theorems 1–2) that guarantees the loop never degrades the open-loop estimate. The recommended configuration is a gated front-end (COP or MUSIC by regime) driving a physics-based, interchangeable SOTA back-end (here δ-GLMB).

For Gaussian or vanishing-cumulant sources the higher-order gain degenerates and the classical $M-1$ limit is recovered. Crucially, *Gaussian noise does not contribute to* (5) since all cumulants of order > 2 vanish for Gaussian processes [16]. This provides two key advantages:

1) **Extended degrees of freedom**: The virtual array $M_v = \rho(M-1)+1$ resolves up to $K_{\max} = \rho(M-1)$ sources—a factor-ρ improvement over the classical $M-1$ limit, and the basis of the underdetermined capability.
2) **Gaussian-noise immunity**: Because all cumulants of order > 2 vanish for Gaussian processes, the cumulant domain is insensitive to additive Gaussian noise of *arbitrary* covariance—including spatially-colored, unknown noise—whereas covariance-based methods require a known or white noise model. The effective cumulant-domain SNR scales as $p_s^\rho/\sigma_c^2 \propto \mathrm{SNR}^\rho$.

The price of these advantages is a higher *estimation variance*: for a given snapshot count, fourth-order cumulants are noisier than the second-order covariance. A central theme of this paper is that the *closed loop offsets exactly this penalty*: the tracker-aided refinement of Section III-B (Theorem 2) provably reduces the cumulant-subspace error by a factor $1/(1+\kappa)$, so the higher-order aperture and noise immunity are obtained at a near-second-order variance. The higher-order front-end and the temporal feedback are thus *complementary*—HOS supplies the degrees of freedom and noise robustness that the loop cannot, while the loop supplies the variance reduction that HOS needs.

For $M = 8$ sensors and $\rho = 2$, this yields $M_v = 15$ virtual elements capable of resolving up to $K_{\max} = 14$ sources—double the classical limit of 7.

Remark (Dual-pathway construction). The same virtual array (4) also arises by combining ρ second-order covariance matrices of *non-stationary* sources [1], attaining the extended aperture $M_v = \rho(M-1)+1$ at second-order estimation variance—substantially lower than direct cumulant estimation. We use direct fourth-order cumulants here for generality; the lower-variance pathway is the natural choice for non-stationary sources and narrows COP's variance gap relative to second-order methods.

III. COP Algorithm Family

A. Subspace COP (Base Algorithm)

The COP algorithm [1] performs DOA estimation by jointly optimizing over the signal and noise subspaces of the virtual cumulant matrix. The number of sources K is assumed known or estimated via information-theoretic criteria such as MDL [24]. Given the eigendecomposition

$$\mathbf{C}_{2\rho} = \mathbf{U}_s \boldsymbol{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_n \boldsymbol{\Lambda}_n \mathbf{U}_n^H \tag{6}$$

where $\mathbf{U}_s \in \mathbb{C}^{M_v \times K}$ and $\mathbf{U}_n \in \mathbb{C}^{M_v \times (M_v - K)}$ span the signal and noise subspaces, respectively, the COP pseudo-spectrum is

$$P_{\mathrm{COP}}(\theta) = \frac{\mathbf{a}_v^H(\theta)\,\mathbf{U}_s\mathbf{U}_s^H\,\mathbf{a}_v(\theta)}{\mathbf{a}_v^H(\theta)\,\mathbf{U}_n\mathbf{U}_n^H\,\mathbf{a}_v(\theta)} \tag{7}$$

which simultaneously maximizes the signal subspace projection (numerator) and minimizes the noise subspace projection (denominator).

Peak detection. The DOA estimates are extracted from $P_{\mathrm{COP}}(\theta)$ via local maximum finding with minimum-separation gating. Specifically, a scan angle θ_i is identified as a *local peak* if

$$P_{\mathrm{COP}}(\theta_i) > P_{\mathrm{COP}}(\theta_{i-1}) \text{ and } P_{\mathrm{COP}}(\theta_i) > P_{\mathrm{COP}}(\theta_{i+1}). \tag{8}$$

All local peaks are ranked by descending amplitude, and the top K peaks are selected subject to a minimum angular separation constraint $|\hat{\theta}_j - \hat{\theta}_k| > \delta_{\min}$ for all previously selected estimates, where $\delta_{\min} = 1^\circ$ prevents multiple detections from the same mainlobe. The resulting DOA estimates $\{\hat{\theta}_k\}_{k=1}^K$ are sorted in ascending order.

B. Temporal COP (T-COP)

For non-stationary scenarios where source DOAs change between radar scans, batch cumulant estimation becomes suboptimal [25]. T-COP introduces temporal accumulation with exponential forgetting:

$$\hat{\mathbf{C}}_{2\rho}^{(n)} = \alpha\, \hat{\mathbf{C}}_{2\rho}^{(n-1)} + (1-\alpha)\, \mathbf{C}_{2\rho}^{(n)} \tag{9}$$

where $\hat{\mathbf{C}}_{2\rho}^{(n)}$ is the accumulated cumulant estimate at scan n, $\mathbf{C}_{2\rho}^{(n)}$ is the instantaneous cumulant from the current scan, and $\alpha \in (0,1)$ is the forgetting factor. With per-scan weights $w_i = (1-\alpha)\alpha^i$, the effective number of snapshots after n scans is the *Kish effective sample size*—the squared sum of the weights divided by the sum of their squares—scaled by the per-scan snapshot count T:

$$\begin{aligned} T_{\text{eff}}^{(n)} &= T\, \frac{\left(\sum_{i=0}^{n-1} w_i\right)^2}{\sum_{i=0}^{n-1} w_i^2} \\ &= T\, \frac{(1-\alpha^n)^2(1+\alpha)}{(1-\alpha)(1-\alpha^{2n})} \xrightarrow[n\to\infty]{} T\, \frac{1+\alpha}{1-\alpha}, \end{aligned} \tag{10}$$

which grows monotonically to a finite steady state. We emphasize that the naive geometric weight sum $T(1-\alpha^n)/(1-\alpha)$ is *not* the effective sample size, since it ignores the unequal squared contributions of the exponential weights; (10) follows directly from $\mathrm{Var}(\hat{\mathbf{C}}_{2\rho}^{(n)}) \propto \sum_i w_i^2$ for the weighted estimator. The T-COP pseudo-spectrum at scan n is computed as

$$P_{\text{T-COP}}^{(n)}(\theta) = \frac{\mathbf{a}_v^H(\theta)\, \hat{\mathbf{U}}_s^{(n)} \hat{\mathbf{U}}_s^{(n)H}\, \mathbf{a}_v(\theta)}{\mathbf{a}_v^H(\theta)\, \hat{\mathbf{U}}_n^{(n)} \hat{\mathbf{U}}_n^{(n)H}\, \mathbf{a}_v(\theta)} \tag{11}$$

where $\hat{\mathbf{U}}_s^{(n)}$ and $\hat{\mathbf{U}}_n^{(n)}$ are the signal and noise subspaces of $\hat{\mathbf{C}}_{2\rho}^{(n)}$.

Tracker-aided subspace refinement. When a multi-target tracker provides predicted DOAs $\{\hat{\theta}_k^{\text{pred}}\}_{k=1}^{\hat{K}}$—the constant-velocity *prediction* $\hat{\theta}_k^{\text{pred}} = \theta_k + \dot{\theta}_k\, \Delta t$ propagated to the current scan, so that the prior stays unbiased even for moving targets—T-COP incorporates this prior information. The prior signal subspace is constructed from the predicted steering vectors:

$$\mathbf{U}_s^{\text{prior}} = \mathrm{orth}\Big(\mathbf{A}_v(\hat{\boldsymbol{\theta}}^{\text{pred}})\Big) \in \mathbb{C}^{M_v \times \hat{K}} \tag{12}$$

and the refined subspace is obtained by an inverse-variance (BLUE) fusion on the Grassmann tangent space [26]. Unlike the gauge-dependent linear interpolation $\mathrm{orth}\big((1-w_p)\mathbf{U}_s + w_p \mathbf{U}_s^{\text{prior}}\big)$—which is *not* a well-defined subspace operation, since $\mathbf{U}_s \mapsto \mathbf{U}_s\mathbf{Q}$ for any $\mathbf{Q} \in \mathcal{U}(K)$ represents the same subspace yet changes the interpolant—the proposed refinement is invariant to the basis gauge. We first align the prior basis to the estimate by the orthogonal Procrustes rotation

$$\mathbf{Q}^\star = \mathbf{V}\mathbf{W}^H, \qquad \mathbf{U}_s^H \mathbf{U}_s^{\text{prior}} = \mathbf{W}\boldsymbol{\Sigma}\mathbf{V}^H \text{ (SVD)}, \tag{13}$$

then combine the estimate and the aligned prior in the normal (tangent) coordinates $\mathbf{H}_{\text{est}}, \mathbf{H}_{\text{pri}}$ of the Grassmannian, reporting the result through the gauge-invariant projector $\mathbf{P}_s = \mathbf{U}_s\mathbf{U}_s^H$:

$$\begin{aligned} \hat{\mathbf{H}} &= \big(\boldsymbol{\Sigma}_{\text{est}}^{-1} + \boldsymbol{\Sigma}_{\text{pri}}^{-1}\big)^{-1} \big(\boldsymbol{\Sigma}_{\text{est}}^{-1}\mathbf{H}_{\text{est}} + \boldsymbol{\Sigma}_{\text{pri}}^{-1}\mathbf{H}_{\text{pri}}\big), \\ \mathbf{P}_s^{\text{refined}} &= \mathrm{retr}(\hat{\mathbf{H}}), \end{aligned} \tag{14}$$

where $\boldsymbol{\Sigma}_{\text{est}}, \boldsymbol{\Sigma}_{\text{pri}}$ are the tangent-space error covariances of the data estimate and the aligned prior, and $\mathrm{retr}(\cdot)$ denotes the Grassmann retraction. The prior covariance is induced from the tracker's predicted-DOA covariance $\mathbf{C}_\theta$ through the steering Jacobian, $\boldsymbol{\Sigma}_{\text{pri}} = \mathbf{G}\mathbf{C}_\theta\mathbf{G}^H$ with columns $\mathbf{G}_k \propto \mathbf{P}_{\mathbf{A}_v}^{\perp}\, \dot{\mathbf{a}}_v(\theta_k)$; in the isotropic special case $\boldsymbol{\Sigma}_{\text{est}} = \sigma_e^2\mathbf{I}$, $\boldsymbol{\Sigma}_{\text{pri}} = \sigma_p^2\mathbf{I}$ this recovers a scalar weight $w_p = \sigma_e^2/(\sigma_e^2+\sigma_p^2)$. The refinement (14) provably reduces the subspace MSE under an explicit, checkable prior-accuracy condition (Theorem 2, Section VI), creating a principled feedback loop in which accurate tracking lowers estimation error and improved estimation in turn sharpens tracking.

Gated operation. In deployment the refinement (14) is guarded by two gates (Algorithm 1), both of which must pass to engage the loop. A *velocity* gate resets the accumulator when the tracker's mean predicted DOA-rate exceeds τ_v and—absent a reliable velocity estimate (e.g. at track initiation)—defaults conservatively to open-loop, so an early accumulation never smears moving sources. A *subspace-bias* gate $d^2 \le \delta_g$ then operationalizes Theorem 2's prior-accuracy condition, with $d^2 = \frac{1}{2\hat{K}}\|\mathbf{U}_s\mathbf{U}_s^H - \mathbf{U}^{\text{prior}}\mathbf{U}^{\text{prior}H}\|_F^2$ the normalized chordal distance between the data and prior subspaces—a checkable surrogate for $\|\mathbf{b}\|^2 < D(\sigma_e^2+\sigma_p^2)$. When either fails the estimator returns *exactly* the open-loop COP estimate, guaranteeing no degradation (deployed $\tau_v = 0.4°$/scan, $\delta_g = 0.3$, $\alpha = 0.85$).

C. Sequential Deflation COP (SD-COP)

SD-COP is a *supporting* extension—not a central contribution—that pushes operation into the super-underdetermined regime $K > \rho(M-1)$; it complements the closed-loop guarantee of Theorem 1, whose corollary shows that sources beyond the static virtual-array limit remain identifiable once the temporal prior makes $\mathbf{J}_{k|k-1} \succ 0$ on the data-unresolved subspace. When K approaches the COP capacity $\rho(M-1)$, batch estimation accuracy degrades due to exhaustion of virtual degrees of freedom. SD-COP addresses this through iterative signal-domain deflation, analogous to successive interference cancellation (SIC) in communications [27]. Rather than deflating in the cumulant domain—which requires accurate estimation of source cumulant powers and is prone to error accumulation—SD-COP operates on the original received signal $\mathbf{X}$ and recomputes the cumulant at each stage from the deflated signal residual.

At each stage, the already-estimated sources are reconstructed by regularized least squares and subtracted from the *original* received signal $\mathbf{X}$—always referencing the cumulative DOA set, so deflation never cascades error—after which COP is re-applied to the cumulant of the residual to discover further sources. Each stage respects the per-stage capacity $K_{\text{batch}} \le \rho(M-1)$, and an optional leave-one-out pass re-estimates each DOA from the residual of all others. As a supporting extension, we omit the (straightforward sequential) pseudocode.

Algorithm 1 Gated T-COP refinement (single scan n)

Require: snapshot $\mathbf{X}_n$; predictions $\hat{\boldsymbol{\theta}}^{\text{pred}}, \dot{\boldsymbol{\theta}}^{\text{pred}}$ ($\varnothing$ if unavailable); thresholds τ_v, δ_g; forgetting α

$\hat{\mathbf{C}} \leftarrow \text{cum}_{2\rho}(\mathbf{X}_n)$, $\hat{K} \leftarrow$ source count
// **Gate 1 (velocity):** accumulation valid only for slow sources
if $\hat{\boldsymbol{\theta}}^{\text{pred}} \neq \varnothing$ **then**
  $m \leftarrow \text{mean}_k\, |\dot{\theta}_k^{\text{pred}}|$
else
  $m \leftarrow \infty$ // no estimate $\Rightarrow$ assume moving
end if
slow $\leftarrow (\hat{\boldsymbol{\theta}}^{\text{pred}} \neq \varnothing) \wedge (m < \tau_v)$
if slow **then**
  $\hat{\mathbf{C}}_{\text{acc}} \leftarrow \alpha\, \hat{\mathbf{C}}_{\text{acc}} + (1-\alpha)\, \hat{\mathbf{C}}$ // accumulate
else
  $\hat{\mathbf{C}}_{\text{acc}} \leftarrow \hat{\mathbf{C}}$ // reset (current scan only)
end if
// **Gate 2 (subspace bias):** enforces Theorem 2 prior-accuracy
prior_ok $\leftarrow$ false
if slow $\wedge$ $(\hat{\boldsymbol{\theta}}^{\text{pred}} \neq \varnothing)$ **then**
  $\mathbf{U}_s \leftarrow$ signal subspace of $\hat{\mathbf{C}}_{\text{acc}}$; $\mathbf{U}^{\text{prior}} \leftarrow \text{orth}(\mathbf{A}_v(\hat{\boldsymbol{\theta}}^{\text{pred}}))$
  $d^2 \leftarrow \frac{1}{2\hat{K}} \|\mathbf{U}_s \mathbf{U}_s^H - \mathbf{U}^{\text{prior}} \mathbf{U}^{\text{prior}H}\|_F^2$
  prior_ok $\leftarrow (d^2 \leq \delta_g)$
end if
if prior_ok **then**
  $P \leftarrow$ prior-refined T-COP spectrum via (14)
  $\hat{\boldsymbol{\theta}} \leftarrow$ prior-guided peaks of P // engage loop
else
  $P \leftarrow$ open-loop COP spectrum of $\hat{\mathbf{C}}_{\text{acc}}$; $\hat{\boldsymbol{\theta}} \leftarrow$ peaks of P // fall back
end if
return $\hat{\boldsymbol{\theta}}$ (and P for tracker likelihoods)

IV. RFS Multi-Target Tracking Back-End

A. Random Finite Set Formulation

The multi-target DOA tracking problem is naturally modeled using random finite sets (RFS) [28], [29]. Extensions such as the cardinalized PHD [30], labeled multi-Bernoulli filter [22], and data association schemes [18], [20] have been proposed. The present work differs by embedding the tracker in a *closed loop* with an interchangeable subspace front-end (COP or MUSIC); the higher-order COP additionally enables underdetermined DOA estimation, and front-end-spectrum-informed birth—capabilities absent from all prior PHD-based DOA tracking pipelines. At scan n, the multi-target state is the RFS

$$\mathcal{X}_n = \{\mathbf{x}_{n,1}, \ldots, \mathbf{x}_{n,K_n}\} \tag{15}$$

where K_n is the (unknown, time-varying) number of targets and each $\mathbf{x}_{n,k} = [\theta_k, \dot{\theta}_k]^T$ contains the DOA and DOA rate. The measurement set from the subspace front-end (COP or MUSIC) is

$$\mathcal{Z}_n = \{z_{n,1}, \ldots, z_{n,M_n}\} \tag{16}$$

where M_n measurements include both target-generated and clutter-generated DOAs.

The GM-PHD filter [17] propagates the first-order moment (intensity function) of the multi-target posterior density, represented as a Gaussian mixture:

$$v_{n|n}(\mathbf{x}) = \sum_{j=1}^{J_{n|n}} w_{n|n}^{(j)}\, \mathcal{N}(\mathbf{x};\, \mathbf{m}_{n|n}^{(j)},\, \mathbf{P}_{n|n}^{(j)}). \tag{17}$$

B. Physics-Based Motion Model (CV / CA)

Each target state $\mathbf{x} = [\theta, \dot{\theta}]^T$ evolves according to the constant-velocity (CV) model [31]:

$$\mathbf{x}_{n+1} = \mathbf{F}\,\mathbf{x}_n + \mathbf{w}_n, \quad \mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \tag{18}$$

where Δt is the inter-scan interval and $\mathbf{w}_n \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is the process noise with

$$\mathbf{Q} = \sigma_w^2 \begin{bmatrix} \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^2/2 & \Delta t \end{bmatrix}. \tag{19}$$

The measurement model is $z_n = \mathbf{H}\,\mathbf{x}_n + v_n$ with $\mathbf{H} = [1,\, 0]$ and $v_n \sim \mathcal{N}(0, \sigma_\theta^2)$. For *maneuvering* targets (non-zero angular curvature) we use the constant-acceleration (CA) model, augmenting the state to $\mathbf{x} = [\theta, \dot{\theta}, \ddot{\theta}]^T$ with $\mathbf{F}_{\text{CA}} = [\,1, \Delta t, \frac{1}{2}\Delta t^2;\, 0, 1, \Delta t;\, 0, 0, 1\,]$, $\mathbf{H} = [1, 0, 0]$, and a white-noise-jerk $\mathbf{Q}$. Throughout, *physics-based* denotes this kinematic (CV or CA) motion model—shared by the TO-PHD and the labeled-RFS back-ends—with CA preferred under maneuver (§VII-E).

C. Recommended Back-End: Physics-Based δ-GLMB

Because the closed loop is back-end-agnostic, any RFS filter that consumes the front-end DOAs and returns a predicted state may serve as the tracker. Our *recommended* back-end is the labeled-RFS δ-GLMB filter [21]—the state-of-the-art identity-aware tracker—realized with the efficient Gibbs joint predict–update truncation of [36] and the physics-based motion of Section IV. The posterior is a weighted set of global hypotheses, each a set of labeled tracks $(\ell, \mathbf{m}, \mathbf{P})$ with cardinality distribution $\rho(n) = \sum_{|I|=n} w_h$. Algorithm 2 summarizes one closed-loop scan: each parent hypothesis predicts its survivors by the CV/CA model and admits a measurement-driven birth per detection; a Gibbs sampler draws the high-weight joint association vectors (each measurement assigned exclusively to a track, a birth, or clutter), each spawning a child hypothesis whose weight is the product of per-track association costs. Hypotheses are merged by label set and pruned, a detection-based confirmation score rejects clutter while letting confirmed tracks coast through occasional front-end misses, and the cardinality-MAP hypothesis is extracted. The labeled hypotheses preserve identity through crossings and, with the CA motion, make this back-end best-in-class under maneuver (Section VII-E).

Algorithm 2 Physics-based δ-GLMB back-end (one closed-loop scan)

Require: hypotheses $\{(w_h, \{(\ell, \mathbf{m}, \mathbf{P})\})\}$; snapshot $\mathbf{X}_n$; motion $(\mathbf{F}, \mathbf{Q})$ (CV or CA)
1: feed confirmed tracks' predictions $\mathbf{HFm}$ to the front-end; $\mathcal{Z}_n \leftarrow$ front-end$(\mathbf{X}_n)$ (gated, Alg. 1)
2: **for** each parent hypothesis h (top H_p by weight) **do**
3: predict survivors $\mathbf{m} \leftarrow \mathbf{Fm}$, $\mathbf{P} \leftarrow \mathbf{FPF}^T + \mathbf{Q}$; add one birth per z_j
4: costs: detect $\propto$ $p_S p_D \mathcal{N}(z_j; \mathbf{Hm}, S)/\kappa$, miss $\propto$ $p_S(1-p_D)$, death $\propto 1-p_S$
5: Gibbs-sample joint assignments γ (z_j exclusive); each spawns a child, $w \leftarrow w_h \prod_i \mathrm{cost}_i(\gamma_i)$
6: **end for**
7: merge children by label set, normalize, prune to top H
8: update confirmation scores (hit $+1$/miss -1, hysteresis)
9: **return** cardinality-MAP hypothesis (confirmed labeled tracks)

D. Lightweight Alternative: Track-Oriented PHD

For compute-constrained deployment the back-end can instead be a *track-oriented PHD* (TO-PHD), a Gaussian-mixture PHD with a predict–identify–update pipeline that avoids hypothesis sampling. The standard GM-PHD processes measurements in a "predict–update" pipeline where every measurement updates every predicted component, creating $\mathcal{O}(J \cdot M)$ updated components per scan. In DOA tracking scenarios with crossing targets, this leads to track coalescence—distinct targets merge into a single track at crossing points.

Its "predict–identify–update" pipeline leverages the motion model (18) to identify which measurement belongs to which track *before* the Bayesian update. The key innovation is the **Hungarian assignment** [32] on the cost matrix

$$C_{jm} = |\hat{\theta}_j^{\mathrm{pred}} - z_m|, \quad \hat{\theta}_j^{\mathrm{pred}} = [\mathbf{F}\,\mathbf{m}_j]_1 \tag{20}$$

where $[\cdot]_1$ denotes the first element (DOA). The matched track j is then updated via the Kalman equations:

$$\mathbf{K}_j = \mathbf{P}_j^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_j^- \mathbf{H}^T + \sigma_\theta^2)^{-1} \tag{21}$$
$$\mathbf{m}_j^+ = \mathbf{m}_j^- + \mathbf{K}_j (z_{m(j)} - \mathbf{H}\mathbf{m}_j^-) \tag{22}$$
$$\mathbf{P}_j^+ = (\mathbf{I} - \mathbf{K}_j \mathbf{H})\mathbf{P}_j^- \tag{23}$$

where each confirmed track uses *only its matched measurement* $z_{m(j)}$, not all measurements as in the standard PHD update.

Relation to the PHD recursion. The matched, single-measurement correction (21)–(23) is *not* the exact GM-PHD update, which sums each component over all measurements [17]. It instead implements a *track-oriented, marginal-association* approximation: the Hungarian step selects the maximum-a-posteriori measurement-to-track assignment, after which each confirmed track is corrected by its assigned measurement alone—the PHD analogue of track-oriented MHT and global-nearest-neighbour association [2]. This trades the exact first-moment recursion for explicit track identity, which is essential in the crossing scenarios of interest. We therefore call the back-end a *track-oriented PHD* (TO-PHD) and reserve "GM-PHD" for the standard recursion used as a baseline; birth and tentative components still receive the standard multi-measurement PHD update, so the filter reduces to the GM-PHD when the association gate is fully opened.

Key design choices:

- **Velocity-gated merge** (Step 6): Two components i, j are merged only if their DOA (position) and DOA-rate (velocity) discrepancies *separately* satisfy
$$\frac{(\theta_i - \theta_j)^2}{[\mathbf{P}_i]_{11}} < \delta_m^2 \text{ and } |\dot{\theta}_i - \dot{\theta}_j| < \delta_v, \tag{24}$$
with $[\mathbf{P}_i]_{11}$ the position variance. Gating position and velocity *separately*—rather than via a full-state Mahalanobis distance, which would *double-count* the velocity already encoded in the state $\mathbf{m} = [\theta, \dot{\theta}]^T$—prevents track coalescence at crossings, where targets share a position but differ in velocity.
- **Birth from the front-end spectrum** (Step 4): birth weight is proportional to the front-end pseudo-spectrum height P (COP or MUSIC) at the unassociated measurement:
$$w_b = w_0 \cdot \frac{P(z_b)}{\max_\theta P(\theta)} \tag{25}$$
ensuring that strong spectral peaks receive higher initial weight for faster confirmation.
- **Differentiated update** (Step 5): Confirmed tracks receive Kalman updates via (21)–(23) only with their matched measurement, while tentative/birth components receive standard PHD multi-measurement updates. This prevents spurious weight growth from unrelated measurements.

V. Cramér–Rao Lower Bound

A. Standard (Covariance-Based) CRLB

Under the stochastic signal model [33], the Fisher Information Matrix (FIM) for DOA estimation from the covariance matrix is

$$\mathbf{F}_{\mathrm{std}} = 2T\,\mathrm{Re}\big\{(\mathbf{D}^H \mathbf{P}_{\mathbf{A}}^\perp \mathbf{D}) \odot (\mathbf{R}_s \mathbf{A}^H \mathbf{R}^{-1} \mathbf{A} \mathbf{R}_s)^T\big\} \tag{26}$$

where $\mathbf{D} = [\frac{\partial \mathbf{a}}{\partial \theta_1}, \ldots, \frac{\partial \mathbf{a}}{\partial \theta_K}]$ is the steering derivative matrix, $\mathbf{P}_{\mathbf{A}}^\perp = \mathbf{I} - \mathbf{A}(\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H$ is the noise subspace projector, $\mathbf{R}_s$ is the source covariance, $\mathbf{R} = \mathbf{A}\mathbf{R}_s\mathbf{A}^H + \sigma^2\mathbf{I}$ is the data covariance, and $\odot$ denotes Hadamard product. The CRLB for the kth DOA is $\mathrm{CRB}(\theta_k) = [\mathbf{F}_{\mathrm{std}}^{-1}]_{kk}$.

B. Higher-Order (Cumulant) CRLB

For DOA estimation from the 2ρth-order cumulant matrix, the FIM uses the virtual array manifold:

$$\mathbf{F}_{\mathrm{COP}} = 2T_{\mathrm{eff}}\,\mathrm{Re}\Big\{(\mathbf{D}_v^H \mathbf{P}_{\mathbf{A}_v}^\perp \mathbf{D}_v) \odot (\mathbf{R}_s^{(v)} \mathbf{A}_v^H \mathbf{R}_c^{-1} \mathbf{A}_v \mathbf{R}_s^{(v)})^T\Big\} \tag{27}$$

where $\mathbf{A}_v$ and $\mathbf{D}_v$ are the virtual steering and derivative matrices of size $M_v \times K$, $\mathbf{R}_s^{(v)} = \mathrm{diag}(p_s^\rho, \ldots, p_s^\rho)$ reflects the ρth-power signal cumulant, and $\mathbf{R}_c = \mathbf{A}_v \mathbf{R}_s^{(v)} \mathbf{A}_v^H + \sigma_c^2 \mathbf{I}_{M_v}$ is the cumulant-domain data "covariance" with effective noise

$\sigma_c^2 \approx C_\rho/T$ (where C_ρ is the cumulant estimation variance factor; $C_2 = 6$ for fourth-order). We stress that (27) is an *approximate* (white-surrogate) bound: the covariance of the estimated cumulants is in general non-white, $\mathrm{Cov}(\mathrm{vec}\,\hat{\mathbf{C}}_{2\rho}) = \mathbf{\Sigma}_c \neq \sigma_c^2\mathbf{I}$, depending on moments up to order 4ρ [16], [23]; setting $\mathbf{\Sigma}_c = \sigma_c^2\mathbf{I}$ gives (27) as a first-order surrogate, not an exact bound. The exact static bound uses the Slepian–Bangs form with the full $\mathbf{\Sigma}_c$, and the bound relevant to the *closed-loop* tracker is the posterior CRB of Theorem 1 (Section VI).

The COP CRLB is lower than the standard CRLB due to two factors:

1) **Extended aperture**: $M_v > M$ provides more virtual elements, sharpening the effective beamwidth.
2) **Gaussian noise elimination**: The effective "SNR" in the cumulant domain scales as $p_s^\rho/\sigma_c^2 \propto \mathrm{SNR}^\rho \cdot T$, providing faster improvement with SNR.

VI. Theoretical Analysis

The closed-loop architecture rests on two theoretical pillars: a posterior Cramér–Rao bound for the *coupled* estimator–tracker (Theorem 1)—a bound for the complete closed-loop framework, not available from a static CRLB—and an optimality guarantee for the T-COP subspace refinement (Theorem 2). A supporting result (Proposition 1) justifies the velocity-gated association at crossings.

A. Closed-Loop Posterior CRB

Let the per-target state $\mathbf{x}_k = [\theta_k, \dot{\theta}_k]^T$ evolve under the linear-Gaussian model (18), and let the scan-k measurement be the estimated cumulant vector $\hat{\mathbf{c}}_k = \mathrm{vec}(\hat{\mathbf{C}}_{2\rho}^{(k)})$ with mean $\mathbf{c}_k(\boldsymbol{\theta}_k) = (\mathbf{A}_v^* \otimes \mathbf{A}_v)\boldsymbol{\gamma}_s$ and covariance $\mathbf{\Sigma}_c$. The cumulant-domain measurement information about $\boldsymbol{\theta}_k$ is the Slepian–Bangs matrix

$$[\mathbf{J}_k^Z]_{ij} = 2\,\mathrm{Re}\Big[\frac{\partial \mathbf{c}_k^H}{\partial \theta_i}\,\mathbf{\Sigma}_c^{-1}\,\frac{\partial \mathbf{c}_k}{\partial \theta_j}\Big], \qquad (28)$$

which reduces to (27) only in the white surrogate $\mathbf{\Sigma}_c = \sigma_c^2\mathbf{I}$.

Theorem 1 (Closed-loop posterior CRB). *For the coupled system* (18), (28)*, the posterior information matrix obeys the Tichavský–Muravchik–Nehorai recursion* [34]

$$\mathbf{J}_k = \underbrace{\big(\mathbf{F}\mathbf{J}_{k-1}^{-1}\mathbf{F}^T + \mathbf{Q}\big)^{-1}}_{\mathbf{J}_{k|k-1}\ (\text{prior})} + \bar{\mathbf{J}}_k^Z, \qquad \bar{\mathbf{J}}_k^Z = \mathbb{E}[\mathbf{J}_k^Z], \qquad (29)$$

and the closed-loop bound $\mathrm{PCRB}_k = \mathbf{J}_k^{-1}$ *dominates the open-loop (per-scan, data-only) bound in the Löwner order,*

$$\mathbf{J}_k^{-1} = (\mathbf{J}_{k|k-1} + \bar{\mathbf{J}}_k^Z)^{-1} \preceq (\bar{\mathbf{J}}_k^Z)^{-1}, \qquad (30)$$

strictly whenever $\mathbf{J}_{k|k-1} \succ 0$*. Moreover, in the underdetermined regime where the data-only information becomes rank-deficient (*$\lambda_{\min}(\bar{\mathbf{J}}_k^Z) \to 0$*), the open-loop bound diverges while the closed-loop bound stays finite,* $\lambda_{\max}(\mathrm{PCRB}_k) \leq 1/\lambda_{\min}(\mathbf{J}_{k|k-1})$.

Proof sketch. Information additivity (29) is the linear-Gaussian specialization of the recursion in [34]. Since $\mathbf{J}_{k|k-1} \succeq 0$, we have $\mathbf{J}_{k|k-1} + \bar{\mathbf{J}}_k^Z \succeq \bar{\mathbf{J}}_k^Z$, and matrix inversion reverses the order, giving (30); when $\bar{\mathbf{J}}_k^Z$ is singular the right-hand side is unbounded while the left remains bounded by the prior information. □

To our knowledge, (29) is the first CRB for the complete closed-loop higher-order-cumulant/RFS tracker. It makes precise *why* the temporal prior restores identifiability beyond the static virtual-array limit, and is validated against Monte-Carlo simulation in Section VII. As a corollary, K sources with $\rho(M-1) < K$ remain trackable provided $\mathbf{J}_{k|k-1} \succ 0$ on the data-unresolved subspace—the regime exploited by SD-COP (Section III-C). The recursion (29) applies verbatim to a covariance front-end (MUSIC) upon replacing the cumulant measurement information $\mathbf{J}_k^Z$ (28) with its covariance-domain counterpart; in the determined regime both front-ends give finite bounds and the lower-variance second-order statistics dominate, consistent with the experiments of Section VII.

B. Optimality of T-COP Subspace Refinement

Theorem 2 (T-COP refinement dominance). *Let the data and Procrustes-aligned prior subspace estimates have unbiased tangent-space errors with covariances* $\mathbf{\Sigma}_{\mathrm{est}} \succ 0$ *and* $\mathbf{\Sigma}_{\mathrm{pri}} \succ 0$*. Then the refined estimate* (14) *is the best linear unbiased estimate on the Grassmann tangent space and satisfies*

$$\mathrm{Cov}(\hat{\mathbf{H}}) = \big(\mathbf{\Sigma}_{\mathrm{est}}^{-1} + \mathbf{\Sigma}_{\mathrm{pri}}^{-1}\big)^{-1} \preceq \mathbf{\Sigma}_{\mathrm{est}}, \qquad (31)$$

with strict improvement whenever $\mathbf{\Sigma}_{\mathrm{pri}}^{-1} \neq 0$*; equivalently* $\mathbb{E}\|\mathbf{P}_s^{\mathrm{refined}} - \mathbf{P}_0\|_F^2 \leq \mathbb{E}\|\mathbf{U}_s\mathbf{U}_s^H - \mathbf{P}_0\|_F^2$*, with isotropic improvement factor* $1/(1+\kappa)$*,* $\kappa = \sigma_e^2/\sigma_p^2$*. If the prior is biased by* $\mathbf{b}$*, the refinement still reduces MSE if and only if*

$$\|\mathbf{b}\|^2 < D\,(\sigma_e^2 + \sigma_p^2), \qquad D = K(M_v - K). \qquad (32)$$

Proof sketch. On the tangent space at the true subspace, both inputs are linear-Gaussian observations of the same point; (14) is their BLUE, whose covariance is the parallel sum (31), dominated by $\mathbf{\Sigma}_{\mathrm{est}}$ because $\mathbf{\Sigma}_{\mathrm{pri}}^{-1} \succeq 0$. The chordal identity $\|\mathbf{P} - \mathbf{P}_0\|_F^2 = 2\|\mathbf{H}\|_F^2$ converts the covariance bound to subspace MSE; the biased case adds $\|(\mathbf{\Sigma}_{\mathrm{est}}^{-1} + \mathbf{\Sigma}_{\mathrm{pri}}^{-1})^{-1}\mathbf{\Sigma}_{\mathrm{pri}}^{-1}\mathbf{b}\|^2$, yielding (32). □

Condition (32) is exactly what the velocity-gated validation of Section IV enforces: predicted priors whose discrepancy exceeds the gate are rejected, so the refinement is guaranteed beneficial in operation. Together, Theorems 1–2 give a *closed-loop guarantee*—the prior supplies information the data alone lacks (Theorem 1) and the refinement injects it without inflating variance (Theorem 2)—so the per-scan estimation error contracts monotonically toward the posterior-CRB floor (29). This replaces a heuristic geometric-rate argument with a bound grounded in the estimator covariance.

C. Crossing Identity Preservation

Proposition 1 (Identity preservation at a crossing). *Consider two targets momentarily sharing a DOA, with velocity difference* $\Delta v = |\dot{\theta}_i - \dot{\theta}_j|$ *and combined prediction-plus-measurement standard deviation* σ_p*. Conditioned on both targets being detected and on clutter-free association, the* *probability that the Hungarian assignment* (20) *preserves track identity across the crossing is*

$$P_{\text{TIPP}} = \Big[\, 1 - Q\Big(\tfrac{\Delta v\,\Delta t}{\sqrt{2}\,\sigma_p}\Big) \Big]\, p_D^2\, (1 - p_{\text{fa}}), \tag{33}$$

where $Q(\cdot)$ *is the Gaussian Q-function,* Δt *the inter-scan interval,* p_D *the detection probability, and* p_{fa} *the per-gate false-association probability.*

Proof sketch. The inertia-predicted positions separate by $\Delta v\,\Delta t$ while the difference of assignment costs is $\mathcal{N}(0, 2\sigma_p^2)$, so the ordering is preserved with probability $1 - Q(\Delta v\,\Delta t/\sqrt{2}\sigma_p)$. The factor $p_D^2(1 - p_{\text{fa}})$ accounts for both targets being detected and neither track being mis-associated to a clutter measurement; setting $p_D = 1$, $p_{\text{fa}} = 0$ recovers the idealized (clutter-free, full-detection) bound. □

The front-end-spectrum birth weight (25) is used as an empirical detection score—stronger spectral peaks initialize tracks with higher weight—and is validated experimentally in Section VII; we do not claim formal Neyman–Pearson optimality for it.

VII. Experimental Results

Unless noted, we use a ULA with $M = 8$ sensors, $d = \lambda/2$, cumulant order $\rho = 2$, a 0.1° scan grid, and a track-oriented PHD back-end with the parameters of Section IV. All metrics are reported with 95% confidence intervals (CIs) over independent Monte-Carlo trials (500 for the headline underdetermined-detection experiment, 20–60 for the ablation and sweep studies, as noted in each caption). We report detection rate P_d (within 3°), the GOSPA metric [35] ($c = 10^\circ$, $p = 2$) with its localization/missed/false decomposition, the *matched-pair* localization RMSE (over GOSPA assignments within the cutoff, which—unlike a naive RMSE that charges a 90° penalty per miss—is a stable localization measure), and the number of label switches.

A. Underdetermined Detection Capability

We first isolate the front-end. Fig. 2 shows that for $K = 10$ sources with $M = 8$ ($K > M - 1 = 7$), the COP-4th pseudo-spectrum (7) resolves all sources while MUSIC and Capon fail. On the full pipeline ($K = 10$ moving targets, 30 scans, SNR= 15 dB), Table I shows that *COP-RFS detects significantly more targets than the MUSIC–PHD cascade* (a MUSIC front-end driving the same TO-PHD back-end, so the comparison isolates the front-end), $P_d = 60.3\% \pm 0.2$ vs. $51.9\% \pm 0.1$ (non-overlapping CIs). MUSIC is structurally capped: running at most $\min(K, M{-}1) = 7$ sources, its detection cannot exceed $(M{-}1)/K = 70\%$, and for $K > M{-}1$ the unmodeled excess sources contaminate the $M \times M$ subspaces, biasing several peaks beyond the 3° gate so that it reaches only 52%. COP removes this cap through the size-M_v=15 virtual aperture (up to 14 DOF); its 60% is limited only by finite-snapshot fourth-order variance and approaches 100% as SNR and T grow. Localization is comparable (3.0° vs. 2.6°) and the overall GOSPA close—honestly, COP's gain is in *detection*, not localization—making this the one regime where the higher-order front-end is decisively advantageous.

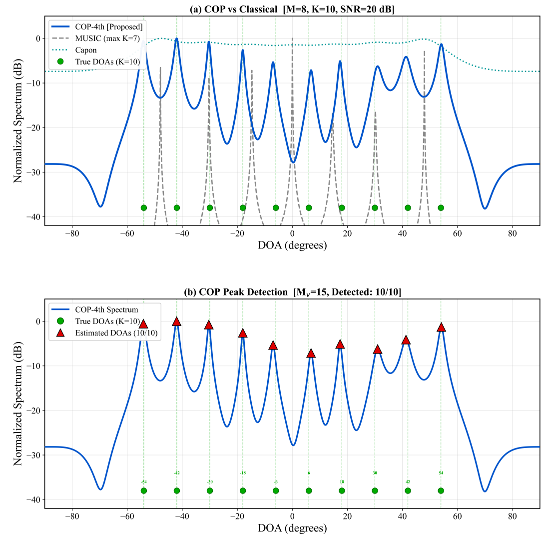

Fig. 2: Spatial spectrum for $K = 10$ sources, $M = 8$ ($K > M{-}1 = 7$): COP-4th resolves all sources via the virtual aperture, while MUSIC and Capon fail. (Top) pseudo-spectra; (bottom) COP peak detection.

TABLE I: Underdetermined end-to-end (K=10 $>$ $M{-}1$=7, 500 trials).

Pipeline	P_d (%)	locRMSE ($^\circ$)	GOSPA ($^\circ$)	Switches
COP-RFS	$\mathbf{60.3 \pm 0.2}$	3.00 ± 0.02	16.69 ± 0.04	187 ± 3
MUSIC-PHD	51.9 ± 0.1	$\mathbf{2.61 \pm 0.01}$	$\mathbf{15.39 \pm 0.01}$	$\mathbf{66 \pm 2}$

Robustness across source count and signal type. The advantage is tied neither to a particular K nor to a particular signal model. Fig. 3(top) sweeps the end-to-end pipeline over $K \in \{2, \ldots, 12\}$ (SNR 12 dB): for $K \le M{-}1$ all methods detect every source, but past K=7 MUSIC collapses toward its $(M{-}1)/K$ cap while COP—and, most strongly, the proposed gated loop—degrade gracefully up to COP's virtual-array limit $\rho(M{-}1) = 14$ (at K=10, $P_d = 97.8\%$ gated, 79.0% open COP, 25.5% MUSIC—the lower MUSIC value than Table I reflecting this sweep's harsher 12 dB, stationary setting). This sharp transition at $M{-}1$ is the structural signature of the extended aperture, not parameter tuning. Fig. 3(bottom) fixes $K = 10 > M{-}1$ and sweeps the source *signal type*: the higher-order front-end sustains detection across structured non-Gaussian radar and communication waveforms (FM 79%, LFM chirp 77%, AM 69%, PSK 63%) and—exactly as the cumulant theory predicts—degrades for purely *Gaussian* sources (32%, whose fourth-order cumulant vanishes) and for severely amplitude-fluctuating Swerling returns (30%), while always exceeding the second-order MUSIC, which cannot surpass its $M{-}1$ ceiling underdetermined. The HOS front-end is thus broadly applicable to non-Gaussian sources, its Gaussian-source limit an explicit and expected boundary.

B. The Closed Loop is Front-End-Agnostic

We next isolate the closed loop via a 2×2 (front-end $\times$ loop) study in the regime where Theorem 2's prior-accuracy condition holds: low SNR (5 dB), slowly varying targets

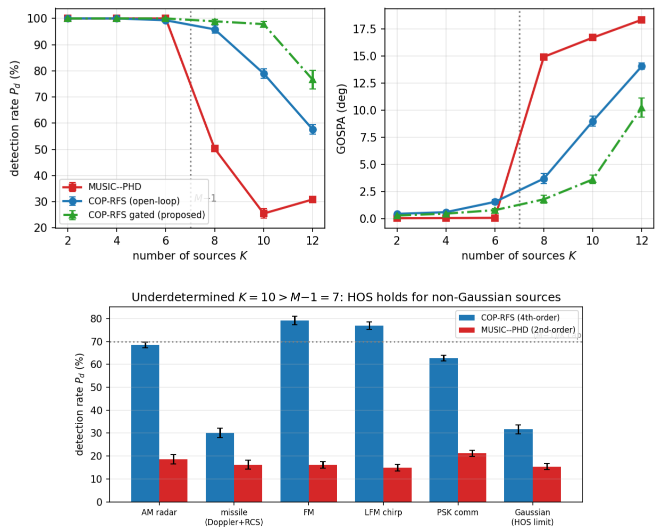

Fig. 3: Robustness sweeps (20 trials, 95% CIs). *Top:* end-to-end detection P_d and GOSPA vs. source count K ($M=8$, near-stationary)—MUSIC collapses past $M-1$=7 while COP and the proposed gated loop degrade gracefully. *Bottom:* P_d vs. source signal type at $K=10>M-1$—the higher-order front-end holds for structured non-Gaussian radar/comm waveforms and degrades, as predicted, for purely Gaussian sources (vanishing 4th cumulant) and severely fluctuating Swerling returns.

TABLE II: 2×2 front-end $\times$ loop (stationary, SNR= 5 dB, $K=6$).

Front-end	open-loop locRMSE	closed-loop locRMSE
COP	0.71 ± 0.02	$\mathbf{0.31 \pm 0.01}$
MUSIC	0.09 ± 0.00	$\mathbf{0.05 \pm 0.00}$

($K=6$). Table II shows the tracker-aided refinement *lowers the localization error of both front-ends*—COP $0.71° \to 0.31°$ (-56%), MUSIC $0.09° \to 0.05°$ (-44%)—confirming the closed loop as a front-end-agnostic principle, exactly as Theorem 2 predicts, not a COP-specific trick. The larger COP gain pays down precisely the fourth-order variance penalty of the higher-order front-end (§II), realizing HOS's extended aperture and Gaussian-noise immunity at a variance approaching second-order MUSIC's—the synergy of HOS made practical *within* a closed loop that is the paper's central message. (In this determined regime the lower-variance MUSIC is the more accurate front-end, consistent with §VII-A.)

This gain persists across the determined regime. Fig. 4 sweeps SNR at $K=6$: the closed loop halves COP's localization error at *every* SNR, while MUSIC's gain is large at low SNR and vanishes as SNR grows—precisely Theorem 2's $1/(1+\kappa)$ behaviour, since the prior helps most when the data subspace is noisy (κ large). COP benefits more at all SNR because its fourth-order variance keeps κ large. Hence the closed loop helps *even where MUSIC is applicable*, and the *deployed* gated variant (GCL) matches the full loop throughout (its velocity gate engages for these slow targets), so the practical system attains the gain. MUSIC is therefore a first-class front-end of the proposed gated loop—indeed the *preferred* one in the determined regime, where its lower variance makes it more accurate than COP—so the recommended system selects the front-end by regime (COP when $K>M-1$, MUSIC otherwise) within the same closed loop.

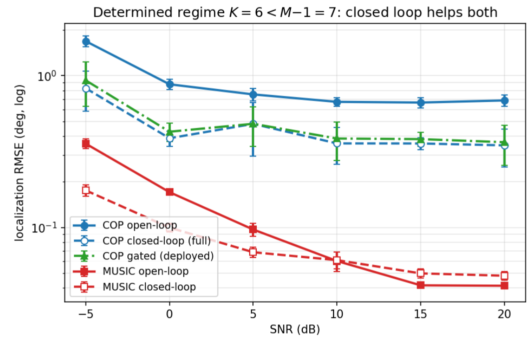

Fig. 4: Determined regime ($K=6<M-1$): localization RMSE vs. SNR for COP and MUSIC (open- and closed-loop) plus the deployed gated closed loop (GCL), which overlaps the full closed loop (30 trials, 95% CIs). The closed loop helps both front-ends—most at low SNR—and helps COP at every SNR because its higher-order variance keeps the prior informative.

C. Robustness to Target Motion

The temporal accumulation that drives the closed-loop gain is valid only for slowly varying targets: for fast targets it smears the moving sources and the accumulating loop *collapses* (on a $K=4$, 2.5°/scan crossing the full closed loop drops P_d from 80.4% to 14.6%; $60.3\% \to 44.6\%$ underdetermined, and MUSIC's loop degrades identically). The gate of Algorithm 1 prevents this: under motion it disables accumulation and feeds back the constant-velocity *prediction* $\theta + \dot{\theta}\,\Delta t$ (so Theorem 2's bias $\|\mathbf{b}\|$ is governed by the prediction error, not the displacement), making the refinement *safe at arbitrary speed*—on the same crossing it matches open-loop COP across SNR$\in\{0,5,15\}$ dB ($P_d = 77.9$–79.4% vs. 78.3–80.4%, no collapse). The deployed system thus accumulates for slow targets (the gain of §VII-B) and falls back otherwise, improving estimation when the prior is informative and never degrading it.

Beyond doing no harm, the motion-compensated loop yields a genuine *identity* benefit for moving targets: on the underdetermined moving scenario ($K=10$), at matched detection ($P_d \approx 52\%$) and localization, the prediction-aligned estimates stabilize data association through crossings, reducing label switches from 214 ± 9 to 181 ± 10 (-15%, non-overlapping CIs). For moving targets the closed loop's value is thus *track continuity*, not per-scan localization (which the tracker's own filtering already optimizes)—complementing its detection gain when underdetermined and its estimation gain at low SNR.

Table III consolidates these results into a single statement of the closed loop's efficacy. The temporal feedback delivers a *concrete, quantified gain in every regime where the prior is informative*: it restores identifiability where the open-loop

TABLE III: Closed-loop efficacy: quantified gain over the open-loop cascade. Each entry is substantiated, with 95% CIs, in the cited theorem/section.

Benefit (where)	Metric	Open-loop	Closed-loop
Identifiability, $K \to M_v - 1$ (Thm 1)	PCRB	diverges	**finite**
Localization, COP @ 5 dB (§VII-B)	locRMSE	0.71°	$\mathbf{0.31}^\circ$ (-56%)
Localization, MUSIC @ 5 dB (§VII-B)	locRMSE	0.09°	$\mathbf{0.05}^\circ$ (-44%)
Track continuity, moving K=10 (§VII-C)	switches	214	**181** (-15%)
Fast motion, safety (§VII-C)	P_d	baseline	**no loss**

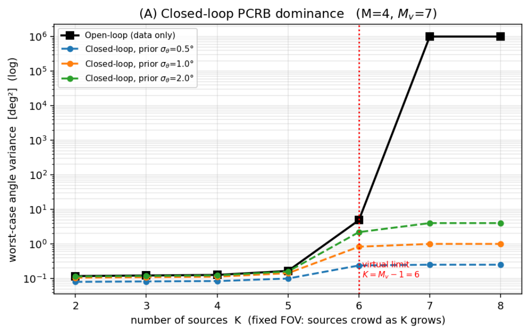

Fig. 5: Validation of Theorem 1: open-loop bound diverges at the virtual-array limit while the closed-loop PCRB stays finite; near-parity in the well-determined regime.

bound diverges (Theorem 1), *halves* COP's low-SNR localization error (-56%), and *cuts moving-target identity switches by* 15%—while the validation gate (Theorem 2) and the motion-compensated prior guarantee it *never degrades* the open-loop baseline where the prior is not. This asymmetry—real gains when the prior helps, provable no-harm when it does not—is the practical case for closing the loop.

D. Validation of the Theory

Closed-loop posterior CRB (Theorem 1). Fig. 5 plots the worst-case angle bound versus source count. The open-loop (data-only) bound diverges as K approaches the virtual-array limit $M_v - 1$, whereas the closed-loop PCRB (29) remains bounded by the prior information $1/\lambda_{\min}(\mathbf{J}_{k|k-1})$, confirming that the temporal prior restores identifiability. In the well-determined regime the two bounds coincide, matching the regime delineation above.

Minimum-variance refinement (Theorem 2). Fig. 6 validates (31)–(32): the fused subspace MSE matches the predicted improvement factor $1/(1+\kappa)$ for an unbiased prior, and the empirical break-even point of the biased case lands at $\|\mathbf{b}\|^2 = D(\sigma_e^2 + \sigma_p^2)$, with the validation gate keeping the fused MSE at or below the data-only MSE.

E. Identity Preservation at Crossings

Finally we evaluate the tracking back-end on the COP front-end ($K = 4$ crossing targets, 30 scans, 40 trials), comparing our track-oriented TO-PHD against the standard GM-PHD [17], the labeled multi-Bernoulli (LMB) filter [22], and the full δ-GLMB filter [21] implemented with the Gibbs-sampling truncation of [36]—the state-of-the-art labeled-RFS trackers requested in review. Table VI shows all three *labeled*

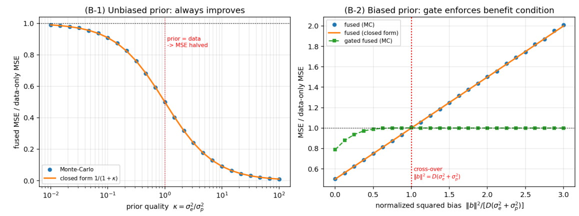

Fig. 6: Validation of Theorem 2: (left) unbiased-prior MSE follows $1/(1+\kappa)$; (right) biased-prior break-even at $\|\mathbf{b}\|^2 = D(\sigma_e^2 + \sigma_p^2)$ and the gated estimator never exceeds the data-only MSE.

TABLE IV: Physics-based motion model for *maneuvering* targets ($K = 4$ sinusoidal maneuver, SNR 6 dB, 30 trials, 95% CIs). The constant-acceleration (CA) physics makes the SOTA δ-GLMB best-in-class on every metric; CV δ-GLMB is misled by its lagging predictions and trails even the CV TO-PHD.

Back-end (motion)	P_d (%)	locRMSE (°)	GOSPA (°)	Switches
TO-PHD (CV)	96.8 ± 0.9	1.41 ± 0.03	3.12 ± 0.10	6.7 ± 2.9
δ-GLMB (CV)	85.9 ± 1.1	1.21 ± 0.07	4.28 ± 0.17	13.1 ± 2.1
δ-GLMB (CA, physics)	$\mathbf{97.5} \pm 1.1$	$\mathbf{0.67 \pm 0.06}$	$\mathbf{1.62} \pm 0.23$	$\mathbf{1.8 \pm 1.0}$

filters reduce identity switches by an order of magnitude over the unlabeled GM-PHD (17–37 vs. 385, whose first-moment recursion coalesces tracks at crossings), at comparable detection (80–82%). The full δ-GLMB—propagating the joint association-history distribution that LMB approximates by its marginals—attains the cleanest identity (17 ± 2 switches) and sharpest localization (locRMSE 0.91°), while LMB attains the best aggregate GOSPA; the small gap confirms LMB as a faithful cheaper approximation, and our TO-PHD a competitive lightweight alternative. The point is that the framework is *agnostic to the back-end*: the closed-loop front-end can drive any RFS tracker, trading computation for identity accuracy. (This determined crossing stresses identity; the COP front-end's advantage lies in the underdetermined regime of §VII-A.)

Physics-based motion makes the SOTA back-end best under maneuver. Back-end agnosticism also exposes the *motion model* as a design choice. For *maneuvering* targets (angular tracks with non-zero curvature, typical of ballistic threats off-boresight), a physics-based constant-acceleration (CA) model decisively beats the constant-velocity (CV) default: on a $K = 4$ sinusoidal-maneuver scenario (SNR 6 dB, Table IV) it cuts δ-GLMB's localization error by 45% $(1.21^\circ \to 0.67^\circ)$ and switches sevenfold $(13.1 \to 1.8)$. Tellingly, *CV* δ*-GLMB* trails even the CV TO-PHD here—its joint association misled by lagging predictions—so the physics model is what makes the SOTA back-end robust under maneuver. Our recommended deployed configuration is therefore the *gated closed-loop higher-order front-end driving a physics-based, interchangeable SOTA back-end* (instantiated with δ-GLMB): agnostic in both estimator and tracker, gated for safety, and matched to the target dynamics.

The front-end loop and the labeled back-end compose

TABLE V: Gated closed-loop front-end × SOTA δ-GLMB back-end (30 trials, 95% CIs). "Ungated" is the best naive loop per regime (full accumulation when slow, motion-compensated when fast). The gate captures the localization gain where the prior is reliable and falls back exactly to open-loop under motion; the ungated loop—lowest error when stationary—is not deployable across regimes.

Regime	Metric	open	ungated	gated
Stationary, 5 dB	locRMSE (°)	0.68	0.34	**0.47**
Crossing, 15 dB	locRMSE (°)	0.90	1.55	**0.90**
Crossing, 15 dB	switches	17.5	27.8	**17.5**

TABLE VI: Tracking back-end ablation on the COP front-end (K=4 crossing, 40 trials, 95% CIs). All three labeled filters cut switches by an order of magnitude (10–23×) vs. GM-PHD; δ-GLMB gives the cleanest identity and sharpest localization, LMB the best aggregate GOSPA.

Back-end	P_d (%)	locRMSE (°)	GOSPA (°)	Switches
Standard GM-PHD [17]	82.3 ± 0.7	1.35 ± 0.08	6.12 ± 0.14	385 ± 8
TO-PHD (ours)	80.4 ± 0.6	1.03 ± 0.09	5.42 ± 0.21	37 ± 6
LMB [22]	81.6 ± 0.8	0.96 ± 0.07	$\mathbf{5.22 \pm 0.17}$	24 ± 3
δ-GLMB [21], [36]	80.6 ± 0.8	$\mathbf{0.91 \pm 0.07}$	5.35 ± 0.17	$\mathbf{17 \pm 2}$

under the gate. Since the loop is front-end-agnostic (§VII-B) and the back-end interchangeable, we confirm the two combine without conflict (Table V). Composition is regime-dependent: in the low-SNR slow regime the loop sharpens δ-GLMB's localization ($0.68^\circ \to 0.47^\circ$), but at a fast crossing an *ungated* loop *degrades* it (1.55° vs. 0.90°), as the prediction-guided front-end and the tracker's association amplify each other's errors. The gate (Algorithm 1) resolves this—engaging only when the prior is reliable—so the gated system gains where the prior helps and falls back *exactly* to open-loop under motion (last column), extending Theorem 2's no-harm guarantee from the front-end to the *complete system*. A summary of the operating regimes is given in Table VII.

VIII. Conclusion

This paper proposed a *gated closed-loop subspace–RFS framework* (Fig. 1) that couples subspace DOA estimation with RFS-based multi-target tracking and, unlike prior open-loop cascades, makes the coupling itself the object of design and analysis. The central message is that the tracker's temporal prediction is information the open-loop pipeline discards, and that feeding it back is provably beneficial *in a characterized regime* rather than uniformly. Concretely (Table III), the closed loop restores identifiability where the data-only bound diverges (Theorem 1), halves COP's low-SNR localization error (-56%), and cuts moving-target identity switches by 15%, while a validation gate and a motion-compensated prior guarantee it never degrades the open-loop baseline.

The key contributions and findings are:

1) **Closed-loop posterior CRB**: Theorem 1 gives the first CRB for the complete coupled estimator–tracker and shows that the temporal prior strictly tightens the bound, restoring identifiability exactly in the underdetermined regime where the data-only bound diverges.

TABLE VII: Operating-regime summary: where each ingredient helps.

Regime	Best front-end	Closed loop
Determined ($K \leq M-1$)	MUSIC (2nd-order)	helps (if prior accurate)
Underdetermined ($K > M-1$)	COP (higher-order)	helps; gate if fast
Low SNR, slow/stationary	either	strongly helps
Fast motion	open-loop	gated off (no harm)

2) **Provably minimum-variance refinement**: Theorem 2 recasts T-COP subspace refinement as a basis-invariant Grassmann inverse-variance fusion and proves it is minimum-variance under an explicit prior-accuracy condition; a validation gate enforces this condition, so the closed loop improves estimation when the prior is accurate and never degrades it otherwise.
3) **Underdetermined capability, honestly bounded**: With Monte-Carlo evaluation (up to 500 trials) and 95% confidence intervals, and against MUSIC, closed-loop-MUSIC, labeled multi-Bernoulli (LMB), and full δ-GLMB baselines, COP-RFS detects significantly more targets than a MUSIC–PHD cascade when $K > M-1$, while the track-oriented PHD back-end preserves identity through crossings; the closed loop is demonstrated to be both front-end- and back-end-agnostic. We explicitly delineate the operating regime—low SNR and slow target motion for the closed-loop estimation gain, $K > M-1$ for the detection gain—rather than claiming uniform superiority; SD-COP is proposed to extend operation toward the super-underdetermined regime.

The recommended deployed system—a gated closed-loop front-end (higher-order COP when underdetermined, covariance MUSIC when determined) driving a physics-based, interchangeable SOTA back-end (here δ-GLMB)—is thus agnostic in both the estimator and the tracker, with the loop gated for safety and the motion model matched to the target dynamics. The proposed framework is particularly relevant to defense, speech enhancement, and communications applications where multiple targets must be simultaneously detected and tracked using compact sensor arrays.

Future directions include: (i) extension to 2-D (azimuth–elevation) DOA with uniform rectangular arrays for full spatial tracking; (ii) environment-adaptive motion models accounting for altitude-dependent atmospheric density and drag coefficients in defense scenarios; and (iii) real-time embedded implementation with operational radar hardware.

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